

```

83.     print(f"Best parameters: short_ma={best_params[0]}, long_ma={best_params[1]}")
84.     print(f"Training Sharpe: {best_sharpe:.4f}")
85.
86.     # Apply best parameters to test data
87.     short_ma, long_ma = best_params
88.     test_data['SMA_Short'] = test_data['Adj Close'].rolling(window=short_ma).mean()
89.     test_data['SMA_Long'] = test_data['Adj Close'].rolling(window=long_ma).mean()
90.
91.     # Generate signals
92.     test_data['Signal'] = 0
93.     test_data.loc[test_data['SMA_Short'] > test_data['SMA_Long'], 'Signal'] = 1
94.
95.     # Run backtest on test data
96.     test_backtest = backtest_strategy(test_data)
97.
98.     # Evaluate test results
99.     test_metrics = evaluate_backtest(test_backtest)
100.
101.     print(f"Test Sharpe: {test_metrics['Sharpe Ratio']:.4f}")
102.
103.     # Store results
104.     result = {
105.         'Train Start': current_start,
106.         'Train End': train_end,
107.         'Test Start': train_end,
108.         'Test End': test_end,
109.         'Short MA': short_ma,
110.         'Long MA': long_ma,
111.         'Train Sharpe': best_sharpe,
112.         'Test Sharpe': test_metrics['Sharpe Ratio'],
113.         'Test Return': test_metrics['Total Return'],
114.         'Test Drawdown': test_metrics['Maximum Drawdown']
115.     }
116.     walk_forward_results.append(result)
117.
118.     # Move to next period
119.     current_start = train_end
120.

```